Deep Generative Models

Dr. Amini

دانشگاه صنعتی شریف

department: Electrical Engineering

Kooshan Fattah Hesari 401102191

Problem Set 1

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1. Properties of Gaussian Distributions

Linear Transformation on Gaussian

If $\vec{X} \in \mathbb{R}^d$ is a gaussian ,$X \sim \mathcal{N}(\vec{\mu}, \Sigma)$, then for a fixed matrix $A_{m\times d}$ and $\vec{b} \in \mathbb{R}^d$ we have :

$$\mathbb{E}[\vec{Y}] = \mathbb{E}[A\vec{X} + \vec{b}] = A\,\mathbb{E}[\vec{X}] + \vec{b} = A\vec{\mu} + \vec{b},$$

$$\mathrm{Cov}(\vec{Y}) = \mathrm{Cov}(A\vec{X} + \vec{b}) = \mathrm{Cov}(A\vec{X}) = A\;\mathrm{Cov}(\vec{X})\,A^\top = A\Sigma A^\top.$$

Now in order to show that the linear transformation of $\vec{X}$ is also gaussian, we use the definition of moment generating functions.

$$\begin{aligned}M_{\vec{Y}}(t) &= \mathbb{E}\big[\exp(t^\top \vec{Y})\big] \\ &= \mathbb{E}\big[\exp(t^\top(A\vec{X} + \vec{b}))\big] = \mathbb{E}\big[\exp(t^\top A\vec{X})\exp(t^\top\vec{b})\big] = \exp(t^\top\vec{b})\mathbb{E}\big[\exp(t^\top A\vec{X})\big] \\ &= \exp(t^\top\vec{b})M_{\vec{X}}(A^\top t)\end{aligned}$$

The moment generating function of a multivariate normal distribution is equal to :

$$M_{\vec{x}}(t) = \exp\big[t^\top\vec{\mu} + \frac{1}{2}t^\top\Sigma t\big]$$

Proof

The moment-generating function of a random vector $\mathbf{X}$ is defined as

$$M_{\mathbf{X}}(\mathbf{t}) \; = \; \mathbb{E}\Big[e^{\mathbf{t}^\top \mathbf{X}}\Big], \qquad \mathbf{t} \in \mathbb{R}^n.$$

Applying the law of the unconscious statistician, we have

$$M_{\mathbf{X}}(\mathbf{t}) = \int_{\mathcal{X}} e^{\mathbf{t}^\top \mathbf{x}} \, f_{\mathbf{X}}(\mathbf{x}) \, d\mathbf{x}.$$

With the probability density function of the multivariate normal distribution, we have

$$M_{\mathbf{X}}(\mathbf{t}) = \int_{\mathbb{R}^n} \exp\big[\mathbf{t}^\top \mathbf{x}\big] \cdot \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \cdot \exp\left[-\frac{1}{2}(\mathbf{x}-\mu)^\top \Sigma^{-1}(\mathbf{x}-\mu)\right] d\mathbf{x}.$$

Now we summarize the two exponential functions inside the integral:

$$\begin{aligned}
&M_{\mathbf{X}}(\mathbf{t}) \\
&= \int_{\mathbb{R}^n} \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \cdot \exp\left[-\frac{1}{2}(\mathbf{x}-\mu)^\top \Sigma^{-1}(\mathbf{x}-\mu) + \mathbf{t}^\top \mathbf{x}\right] d\mathbf{x} \\
&= \int_{\mathbb{R}^n} \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \cdot \exp\left[-\frac{1}{2}\big(\mathbf{x}^\top \Sigma^{-1}\mathbf{x} - 2\mu^\top \Sigma^{-1}\mathbf{x} + \mu^\top \Sigma^{-1}\mu - 2\mathbf{t}^\top \mathbf{x}\big)\right] d\mathbf{x} \\
&= \int_{\mathbb{R}^n} \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \cdot \exp\left[-\frac{1}{2}\big(\mathbf{x}^\top \Sigma^{-1}\mathbf{x} - 2(\mu + \Sigma\mathbf{t})^\top \Sigma^{-1}\mathbf{x} + \mu^\top \Sigma^{-1}\mu\big)\right] d\mathbf{x} \\
&= \int_{\mathbb{R}^n} \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \cdot \exp\left[-\frac{1}{2}\Big((\mathbf{x} - (\mu + \Sigma\mathbf{t}))^\top \Sigma^{-1}(\mathbf{x} - (\mu + \Sigma\mathbf{t})) - 2\mathbf{t}^\top \mu + \mathbf{t}^\top \Sigma \mathbf{t}\Big)\right] d\mathbf{x} \\
&= \exp\big[\mathbf{t}^\top \mu + \tfrac{1}{2}\mathbf{t}^\top \Sigma \mathbf{t}\big] \int_{\mathbb{R}^n} \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \cdot \exp\left[-\frac{1}{2}(\mathbf{x} - (\mu + \Sigma\mathbf{t}))^\top \Sigma^{-1}(\mathbf{x} - (\mu + \Sigma\mathbf{t}))\right] d\mathbf{x}.
\end{aligned}$$

The integrand is equal to the probability density function of a multivariate normal distribution:

$$M_{\mathbf{X}}(\mathbf{t}) = \exp\big[\mathbf{t}^\top \mu + \tfrac{1}{2}\mathbf{t}^\top \Sigma \mathbf{t}\big] \int_{\mathbb{R}^n} \mathcal{N}(\mathbf{x};\, \mu + \Sigma\mathbf{t},\, \Sigma) \, d\mathbf{x}.$$

Because the entire probability density integrates to one, we finally have:

$$M_{\mathbf{X}}(\mathbf{t}) = \exp\big[\mathbf{t}^\top \mu + \tfrac{1}{2}\mathbf{t}^\top \Sigma \mathbf{t}\big] \, .$$

therefore we have :

$$\begin{aligned}
M_{\vec{Y}}(t) = \exp(t^\top \vec{b}) M_{\vec{X}(A^\top t)} &= \exp(t^\top \vec{b}) \exp\left\lceil t^\top A\vec{\mu} + \frac{1}{2} t^\top A \Sigma A^\top t \right\rceil \\
&= \exp\left\lceil t^\top (A\vec{\mu} + \vec{b}) + \frac{1}{2} t^\top (A \Sigma A^\top) t \right\rceil
\end{aligned}$$

Because moment-generating function and probability density function of a random variable are equivalent, this shows that $\vec{Y}$ is following a multivariate normal distribution with variance $A\Sigma A^\top$ and mean $A\vec{\mu} + \vec{b}$.

$$\vec{Y} \sim \mathcal{N}(\vec{\mu}_y, \Sigma_y) \quad ; \mu_y = A\vec{\mu} + \vec{b}, \Sigma_y = A\Sigma A^\top$$

Sum of Independent Gaussians

First of all for $a\vec{X}$ and $b\vec{Y}$ we have :

$$\vec{Y}' = b\vec{Y} \to \mathbb{E}\lceil \vec{Y}' \rceil = b\vec{\mu}_Y, \, \Sigma_{Y'} = b^2\Sigma_Y$$
$$\vec{X}' = a\vec{X} \to \mathbb{E}\lceil \vec{X}' \rceil = a\vec{\mu}_X, \, \Sigma_{X'} = a^2\Sigma_X$$

now for $\vec{P}(\vec{Z})$, where $\vec{Z} = a\vec{X} + b\vec{Y}$, we have:

$$\begin{aligned}\vec{P}(\vec{Z}) &= \int_{-\infty}^{\infty} \vec{P}_{x'}(\vec{X}')\vec{P}_{Y'}(\vec{Z} - \vec{X}')d\vec{X}' \\ &= \vec{P}(\vec{X}') * \vec{P}(\vec{Y}')\end{aligned}$$

$$\begin{aligned}M_{\vec{Z}(t)} &= \mathbb{E}_{\vec{Z} \sim \mathbf{P}_{\vec{Z}}}\lceil \exp(t^\top \vec{Z}) \rceil \\ &= M_{\vec{X}'(t) \cdot M_{\vec{Y}'(t)}} \\ &= \exp(t^\top \mu_{X'} + \frac{1}{2}t^\top \Sigma_{X'} t) \exp(t^\top \mu_{Y'} + \frac{1}{2}t^\top \Sigma_{Y'} t) \\ &= \exp(t^\top(\mu_{X'} + \mu_{Y'}) + \frac{1}{2}t^\top(\Sigma_{X'} + \Sigma_{Y'})t)\end{aligned}$$

Therefore the mean of the r.v. $\vec{Z}$ is $\mu_{X'} + \mu_{Y'} = a\vec{\mu}_X + b\vec{\mu}_Y$ and the variance is $\Sigma_{X'} + \Sigma_{Y'} = a^2\Sigma_X + b^2\Sigma_Y$.

Conditional Gaussian

If $\vec{X} \in \mathbb{R}^{d_x}$ and $\vec{Y} \in \mathbb{R}^{d_y}$ for the distribution of the r.v. $\vec{Y}$ where $\vec{}$by using the Moment Generating Function we have :

$$\vec{X} \sim \mathcal{N}(\mu_X, \Sigma_{XX}), \qquad (\vec{Y} \mid \vec{X}) \sim \mathcal{N}(A\vec{X} + b, \Sigma_{Y|X}), \quad A \in \mathbb{R}^{d_y \times d_x}, \; b \in \mathbb{R}^{d_y}.$$

For $Z \sim \mathcal{N}(\mu, \Sigma)$: $\quad M_Z(t) = \mathbb{E}\left[e^{t^\top Z}\right] = \exp\left(t^\top \mu + \frac{1}{2}\, t^\top \Sigma t\right)$.

$$M_{\vec{Y}|\vec{X}=x}(t) = \mathbb{E}\left[e^{t^\top \vec{Y}} \mid \vec{X} = x\right] = \exp\left(t^\top(Ax + b) + \tfrac{1}{2}\, t^\top \Sigma_{Y|X} t\right).$$

$$M_{\vec{Y}}(t) = \mathbb{E}\left[M_{\vec{Y}|\vec{X}}(t)\right] = \exp\left(t^\top b + \tfrac{1}{2}\, t^\top \Sigma_{Y|X} t\right)\, \mathbb{E}\left[e^{t^\top A\vec{X}}\right] = \exp\left(t^\top b + \tfrac{1}{2}\, t^\top \Sigma_{Y|X} t\right)\, M_{\vec{X}}(A^\top t).$$

$$M_{\vec{X}}(A^\top t) = \exp\left(t^\top A\mu_X + \tfrac{1}{2}\, t^\top (A\Sigma_{XX}A^\top) t\right).$$

$$\Rightarrow \; M_{\vec{Y}}(t) = \exp\left(t^\top(A\mu_X + b) + \tfrac{1}{2}\, t^\top \left(A\Sigma_{XX}A^\top + \Sigma_{Y|X}\right) t\right).$$

$$\vec{Y} \sim \mathcal{N}\left(A\mu_X + b,\; A\Sigma_{XX}A^\top + \Sigma_{Y|X}\right).$$

$$\vec{Y} \sim \mathcal{N}\left(A\mu_X + b,\; A\Sigma_{XX}A^\top + \Sigma_{Y|X}\right)$$

General Joint Gaussian Conditional Distribution

For the derivation of the conditional Gaussian distribution, as said the questions, we assume an invertible covariance matrix. let's say that $\vec{X} \in \mathbb{R}^k$ and $\vec{Y} \in \mathbb{R}^l$ We have :

$$\begin{aligned} p(\vec{Y}|\vec{X}) &= \frac{p(\vec{Y},\vec{X})}{p(\vec{X})} \\ &= \frac{\frac{1}{(2\pi)^{\frac{k+l}{2}}}\frac{1}{\det^{\frac{1}{2}}\Sigma}\exp\left[-\frac{1}{2}\left(\begin{bmatrix}\vec{X}-\vec{\mu}_X\\ \vec{Y}-\vec{\mu}_Y\end{bmatrix}\right)^\top \Sigma^{-1}\left(\begin{bmatrix}\vec{X}-\vec{\mu}_X\\ \vec{Y}-\vec{\mu}_Y\end{bmatrix}\right)\right]}{\frac{1}{(2\pi)^{\frac{k}{2}}}\frac{1}{\det^{\frac{1}{2}}\Sigma_{XX}}\exp\left[-\frac{1}{2}\left(\begin{bmatrix}\vec{X}-\vec{\mu}_X\end{bmatrix}\right)^\top \Sigma_{XX}^{-1}\left(\begin{bmatrix}\vec{X}-\vec{\mu}_X\end{bmatrix}\right)\right]} \end{aligned}$$

That $p(\vec{X})$ can be written in this way or that $\vec{X}$ is also gaussian with the given mean and variance follows from the fact that $\vec{X}$and $\vec{Y}$ are jointly gaussian.
We now have to examine the determinant of the partitioned Covariance matrix. Since the determinant of a partitioned matrix can be evaluated as :

$$\det\begin{bmatrix}A_{11} & A_{12}\\ A_{21} & A_{22}\end{bmatrix} = \det(A_{11})\det(A_{22}-A_{21}A_{11}^{-1}A_{12})^*$$

block dagonalization

the fact that we can compute the determinant through the previous identity is a consequence of the block dagonalization identity:

$$\begin{bmatrix}A & B\\ C & D\end{bmatrix} = \begin{bmatrix}I & 0\\ C & I\end{bmatrix}\begin{bmatrix}A & 0\\ 0 & S\end{bmatrix}\begin{bmatrix}I & B\\ 0 & I\end{bmatrix}$$

$$\det\begin{bmatrix}A & B\\ C & D\end{bmatrix} = \det\begin{bmatrix}A & 0\\ 0 & S\end{bmatrix} = \det A\cdot\det S; \quad \text{where S is the Schur Complement wrt A}$$

$$\det A\cdot\det S = \det A\cdot\det(D-CA^{-1}B)$$

It follows that :

$$\det\Sigma = \det\Sigma_{XX}\det(\Sigma_{YY}-\Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY})$$

$$\frac{\det\Sigma}{\det\Sigma_{XX}} = \det(\Sigma_{YY}-\Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY})$$

Therefore

$$p(\vec{Y}|\vec{X}) = \frac{1}{(2\pi)^{\frac{l}{2}}}\frac{1}{\det^{\frac{1}{2}}(\Sigma_{YY}-\Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY})}\exp(-\frac{1}{2}Q)$$

$$Q = \begin{bmatrix}\vec{X}-\vec{\mu}_X\\ \vec{Y}-\vec{\mu}_Y\end{bmatrix}^\top \Sigma^{-1}\begin{bmatrix}\vec{X}-\vec{\mu}_X\\ \vec{Y}-\vec{\mu}_Y\end{bmatrix} - \begin{bmatrix}\vec{X}-\vec{\mu}_X\end{bmatrix}^\top\Sigma_{XX}^{-1}\begin{bmatrix}\vec{X}-\vec{\mu}_X\end{bmatrix}$$

to evaluation Q we use the matrix inversion formula for a partitioned symmetric matrix:

$$\begin{bmatrix}A_{11} & A_{12}\\ A_{21} & A_{22}\end{bmatrix}^{-1} = \begin{bmatrix}\left(A_{11}-A_{12}A_{22}^{-1}A_{21}\right)^{-1} & -A_{11}^{-1}A_{12}\left(A_{22}-A_{21}A_{11}^{-1}A_{12}\right)^{-1}\\ -\left(A_{22}-A_{21}A_{11}^{-1}A_{12}\right)^{-1}A_{21}A_{11}^{-1} & \left(A_{22}-A_{21}A_{11}^{-1}A_{12}\right)^{-1}\end{bmatrix}.$$

Using this form, the off-diagonal blocks are transposes of each other so that the inverse matrix must be symmetric. This is because Σ is symmetric, and hence Σ^{-1} is symmetric. By using the matrix inversion lemma we have :

$$(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} = A_{11}^{-1} + A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1}A_{21}A_{11}^{-1}$$

so that :

$$\Sigma^{-1} = \begin{bmatrix} \Sigma_{xx}^{-1} + \Sigma_{xx}^{-1}\Sigma_{xy}B^{-1}\Sigma_{yx}\Sigma_{xx}^{-1} & -\Sigma_{xx}^{-1}\Sigma_{xy}B^{-1} \\ -B^{-1}\Sigma_{yx}\Sigma_{xx}^{-1} & B^{-1} \end{bmatrix}$$

where :

$$B = \Sigma_{yy} - \Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{xy}.$$

Now by using the block dagonalization identity we have :

$$\Sigma^{-1} = \begin{bmatrix} I & -\Sigma_{xx}^{-1}\Sigma_{xy} \\ 0 & I \end{bmatrix} \begin{bmatrix} \Sigma_{xx}^{-1} & 0 \\ 0 & B^{-1} \end{bmatrix} \begin{bmatrix} I & 0 \\ -\Sigma_{yx}\Sigma_{xx}^{-1} & I \end{bmatrix}$$

and if we set $\tilde{X} = \vec{X} - \vec{\mu}_x$ and $\tilde{Y} = \vec{Y} - \vec{\mu}_Y$:

$$
\begin{aligned}
Q &= \begin{bmatrix} \tilde{\vec{X}} \\ \tilde{\vec{y}} \end{bmatrix}^\top \begin{bmatrix} I & -\Sigma_{XX}^{-1}\Sigma_{XY} \\ 0 & I \end{bmatrix} \begin{bmatrix} \Sigma_{XX}^{-1} & 0 \\ 0 & B^{-1} \end{bmatrix} \begin{bmatrix} I & 0 \\ -\Sigma_{YX}\Sigma_{XX}^{-1} & I \end{bmatrix} \begin{bmatrix} \tilde{\vec{X}} \\ \tilde{\vec{y}} \end{bmatrix} - \tilde{X}^\top \Sigma_{XX}^{-1} \tilde{X} \\
&= \begin{bmatrix} \tilde{\vec{X}} \\ \tilde{\vec{Y}} - \Sigma_{YX}\Sigma_{XX}^{-1}\tilde{\vec{X}} \end{bmatrix}^\top \begin{bmatrix} \Sigma_{XX}^{-1} & 0 \\ 0 & B^{-1} \end{bmatrix} \begin{bmatrix} \tilde{\vec{X}} \\ \tilde{\vec{Y}} - \Sigma_{YX}\Sigma_{XX}^{-1}\tilde{\vec{X}} \end{bmatrix} - \tilde{X}^\top \Sigma_{XX}^{-1} \tilde{X} \\
&= \begin{bmatrix} \tilde{\vec{Y}} - \Sigma_{YX}\Sigma_{XX}^{-1}\tilde{\vec{X}} \end{bmatrix}^\top B^{-1} \begin{bmatrix} \tilde{\vec{Y}} - \Sigma_{YX}\Sigma_{XX}^{-1}\tilde{\vec{X}} \end{bmatrix}
\end{aligned}
$$

and finally we have :

$$
Q = \left[\vec{Y} - \left(E(\vec{Y}) + \Sigma_{YX}\Sigma_{XX}^{-1}(\vec{X} - E(\vec{X}))\right)\right]^\top \left[\Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY}\right]^{-1} \left[\vec{Y} - \left(E(\vec{Y}) + \Sigma_{YX}\Sigma_{XX}^{-1}(\vec{X} - E(\vec{X}))\right)\right]
$$

so that :

$$
\vec{\mu}_{Y|X} = E(\vec{Y}) + \Sigma_{YX}\Sigma_{XX}^{-1}(\vec{X} - E\lceil\vec{X}\rceil)
$$
$$
\Sigma_{Y|X} = \Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY}
$$

KL Divergence between Two Multivariate Gaussians

Let $p = \mathcal{N}(\mu_0, \Sigma_0)$ and $q = \mathcal{N}(\mu_1, \Sigma_1)$ on $\mathbb{R}^d$ with $\Sigma_0, \Sigma_1 \succ 0$. The KL divergence is

$$
D_{\mathrm{KL}}(p\|q) = \mathbb{E}_{X\sim p}\left[\log \frac{p(X)}{q(X)}\right].
$$

Using the Gaussian log–density

$$
\log \mathcal{N}(x \mid \mu, \Sigma) = -\tfrac{1}{2}\left(d \log 2\pi + \log\det\Sigma + (x-\mu)^\top \Sigma^{-1}(x-\mu)\right),
$$

the terms $d \log 2\pi$ cancel in the ratio and we obtain

$$
\log \frac{p(x)}{q(x)} = \tfrac{1}{2}(\log\det\Sigma_1 - \log\det\Sigma_0) + \tfrac{1}{2}\left[(x-\mu_1)^\top \Sigma_1^{-1}(x-\mu_1) - (x-\mu_0)^\top \Sigma_0^{-1}(x-\mu_0)\right].
$$

Taking expectation with respect to $X \sim \mathcal{N}(\mu_0, \Sigma_0)$ reduces the problem to two quadratic expectations. For any symmetric A and any a,

$$
\mathbb{E}\left[(X-a)^\top A (X-a)\right] = \mathrm{tr}(A\Sigma_0) + (a-\mu_0)^\top A (a-\mu_0),
$$

because $X = \mu_0 + (X - \mu_0)$, the cross term has zero mean, and $\mathbb{E}[(X-\mu_0)(X-\mu_0)^\top] = \Sigma_0$. Applying this identity twice gives

$$
\mathbb{E}_p\left[(X-\mu_0)^\top \Sigma_0^{-1}(X-\mu_0)\right] = \mathrm{tr}(\Sigma_0^{-1}\Sigma_0) = d,
$$

$$\mathbb{E}_p\big[(X-\mu_1)^\top\Sigma_1^{-1}(X-\mu_1)\big] = \mathrm{tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1-\mu_0)^\top\Sigma_1^{-1}(\mu_1-\mu_0).$$

Substituting back yields the closed form

$$D_{\mathrm{KL}}(\mathcal{N}(\mu_0,\Sigma_0)\,\|\,\mathcal{N}(\mu_1,\Sigma_1)) = \frac{1}{2}\left[\mathrm{tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1-\mu_0)^\top\Sigma_1^{-1}(\mu_1-\mu_0) - d + \log\frac{\det\Sigma_1}{\det\Sigma_0}\right]$$

For $d=1$ (variances σ_0^2,σ_1^2), the trace and determinants collapse, giving

$$D_{\mathrm{KL}}\big(\mathcal{N}(\mu_0,\sigma_0^2)\,\|\,\mathcal{N}(\mu_1,\sigma_1^2)\big) = \frac{1}{2}\left[\frac{\sigma_0^2}{\sigma_1^2} + \frac{(\mu_1-\mu_0)^2}{\sigma_1^2} - 1 + \log\frac{\sigma_1^2}{\sigma_0^2}\right]$$

2. Auto regressive (AR) Models of Order p

Setup and Notation (assumptions stated clearly)

We work with the AR(p) model

$$y_t \;=\; \phi_1 y_{t-1} \;+\; \phi_2 y_{t-2} \;+\; \cdots \;+\; \phi_p y_{t-p} \;+\; \varepsilon_t,$$
$$\varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\sigma^2), \quad \varepsilon_t \;\perp\; \{y_s : s<t\}.$$

Assumption for likelihood: we use the *conditional* (innovations) likelihood, i.e., we condition on the first p observations $(y_1,\ldots,y_p)$. Define, for $t=p+1,\ldots,n$ and $m := n-p$,

$$\mathbf{y} := [\,y_{p+1},\ldots,y_n\,]^\top, \qquad \boldsymbol{\phi} := [\,\phi_1,\ldots,\phi_p\,]^\top,$$
$$\mathbf{X} := \begin{bmatrix} y_p & y_{p-1} & \cdots & y_1 \\ y_{p+1} & y_p & \cdots & y_2 \\ \vdots & \vdots & \ddots & \vdots \\ y_{n-1} & y_{n-2} & \cdots & y_{n-p} \end{bmatrix}, \qquad \mathbf{y} \;=\; \mathbf{X}\boldsymbol{\phi} \;+\; \boldsymbol{\varepsilon},$$
$$\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0},\ \sigma^2\mathbf{I}_m).$$

Model + data layout: $y_t = \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t$, $\varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\sigma^2)$. Usable sample $t = p+1,\ldots,n$ with $m=n-p$: $\mathbf{y} = \mathbf{X}\boldsymbol{\phi} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0},\sigma^2\mathbf{I}_m)$.

2.1 Likelihood Function

For each $t=p+1,\ldots,n$,

$$y_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}\Big(\sum_{i=1}^p \phi_i y_{t-i}\ ,\ \sigma^2\Big).$$

Therefore, the conditional likelihood (given $y_1,\ldots,y_p$) factorizes as

$$L(\boldsymbol{\phi},\sigma^2 \mid \mathbf{y}_{1:n}) = \prod_{t=p+1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}}\,\exp\left\{-\frac{1}{2\sigma^2}\Big(y_t - \sum_{i=1}^p \phi_i y_{t-i}\Big)^2\right\}.$$

In matrix form,

$$L(\boldsymbol{\phi}, \sigma^2 \mid \mathbf{y}_{1:n}) = (2\pi\sigma^2)^{-m/2} \exp\left\{ -\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\phi})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\phi}) \right\},$$
$$\ell(\boldsymbol{\phi}, \sigma^2) = -\frac{m}{2}\log(2\pi) - \frac{m}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\phi})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\phi}).$$

$$L(\boldsymbol{\phi}, \sigma^2 \mid \mathbf{y}_{1:n}) = (2\pi\sigma^2)^{-m/2} \exp\left\{ -\tfrac{1}{2\sigma^2}\|\mathbf{y} - \mathbf{X}\boldsymbol{\phi}\|_2^2 \right\},$$
$$\ell(\boldsymbol{\phi}, \sigma^2) = -\tfrac{m}{2}\log(2\pi) - \tfrac{m}{2}\log\sigma^2 - \tfrac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\phi})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\phi}).$$

2.2 Maximum Likelihood Estimation (MLE)

Differentiate the log-likelihood w.r.t. $\boldsymbol{\phi}$ and set to zero (treat σ^2 as fixed):

$$\frac{\partial \ell}{\partial \boldsymbol{\phi}} = \frac{1}{\sigma^2}\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\phi}) = \mathbf{0},$$
$$\Longrightarrow \mathbf{X}^\top\mathbf{X}\,\widehat{\boldsymbol{\phi}} = \mathbf{X}^\top\mathbf{y},$$
$$\boxed{\widehat{\boldsymbol{\phi}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}} \quad \text{(assume } \mathbf{X}^\top\mathbf{X} \text{ invertible, i.e., no exact collinearity).}$$

Differentiate w.r.t. σ^2 and set to zero, then plug $\widehat{\boldsymbol{\phi}}$:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{m}{2}\frac{1}{\sigma^2} + \frac{1}{2(\sigma^2)^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\phi})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\phi}) = 0,$$
$$\Longrightarrow \boxed{\widehat{\sigma}^2_{\text{MLE}} = \frac{1}{m}(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\phi}})^\top(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\phi}})} = \frac{1}{n-p}\sum_{t=p+1}^{n}\Big(y_t - \sum_{i=1}^{p}\widehat{\phi}_i y_{t-i}\Big)^2.$$

The maximized log-likelihood (handy for AIC/BIC) is

$$\boxed{\ell(\widehat{\boldsymbol{\phi}}, \widehat{\sigma}^2) = -\frac{m}{2}\log(2\pi) - \frac{m}{2} - \frac{m}{2}\log\widehat{\sigma}^2}.$$

$$\widehat{\boldsymbol{\phi}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y},$$
$$\widehat{\sigma}^2_{\text{MLE}} = \frac{1}{n-p}\sum_{t=p+1}^{n}\Big(y_t - \sum_{i=1}^{p}\widehat{\phi}_i y_{t-i}\Big)^2,$$
$$\ell(\widehat{\boldsymbol{\phi}}, \widehat{\sigma}^2) = -\tfrac{m}{2}\log(2\pi) - \tfrac{m}{2} - \tfrac{m}{2}\log\widehat{\sigma}^2, \quad m = n - p.$$

2.3 Linking AR(1) and the Markov Property — (expanded parts (a) and (b))

We study the AR(1) process

$$y_t = \phi\, y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \qquad \varepsilon_t \perp \mathcal{F}_{t-1}.$$

(a) One–step transition density $p(y_t \mid y_{t-1})$ and Markov property.

Affine–Gaussian argument. Conditional on y_{t-1}, the variable y_t is an affine transformation of ε_t:

$$y_t \;=\; \phi\, y_{t-1} \;+\; \varepsilon_t \quad\Longrightarrow\quad y_t \mid y_{t-1} \;=\; \phi\, y_{t-1} \;+\; \varepsilon_t.$$

Since $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ and is independent of y_{t-1},

$$y_t \mid y_{t-1} \;\sim\; \mathcal{N}\Big(\;\phi\, y_{t-1}\;,\;\sigma^2\;\Big).$$

Thus the transition density (time–homogeneous kernel) is

$$p(y_t \mid y_{t-1}) \;=\; \frac{1}{\sqrt{2\pi\sigma^2}}\;\exp\Bigg\{-\frac{1}{2\sigma^2}\left(y_t - \phi\, y_{t-1}\right)^2\Bigg\}.$$

Why this implies Markov(1). Let A be any Borel set and let $\mathcal{F}_{t-1} = \sigma(y_{t-1}, y_{t-2}, \ldots)$. Using independence of ε_t from $\mathcal{F}_{t-1}$,

$$\mathbb{P}(y_t \in A \mid \mathcal{F}_{t-1}) \;=\; \mathbb{P}\big(\phi y_{t-1} + \varepsilon_t \in A \mid \mathcal{F}_{t-1}\big)$$
$$=\; \mathbb{P}\big(\phi y_{t-1} + \varepsilon_t \in A \mid y_{t-1}\big) \;=\; \int_A p(z \mid y_{t-1})\, dz.$$

Hence

$$\mathbb{P}(y_t \in A \mid y_{t-1}, y_{t-2}, \ldots) \;=\; \mathbb{P}(y_t \in A \mid y_{t-1}),$$

which is precisely the (time–homogeneous) Markov property of order 1.

Optional check via moments. Using the kernel above,

$$\mathbb{E}[y_t \mid y_{t-1}] \;=\; \phi\, y_{t-1}, \qquad \mathrm{Var}(y_t \mid y_{t-1}) \;=\; \sigma^2,$$

both depending on the past *only* through y_{t-1}.

(b) Stationary distribution, its mean/variance, and invariance (assume $|\phi| < 1$).

Causal solution and Gaussianity. Iterating the recursion,

$$y_t \;=\; \phi^k y_{t-k} \;+\; \sum_{j=0}^{k-1} \phi^j\, \varepsilon_{t-j}.$$

If $|\phi| < 1$, then $\phi^k y_{t-k} \to 0$ in mean square as $k \to \infty$ (contractive term), and the infinite sum converges in L^2:

$$y_t \;=\; \sum_{j=0}^{\infty} \phi^j\, \varepsilon_{t-j}.$$

This is a linear combination of i.i.d. Gaussians, hence y_t is Gaussian.

Mean and variance under stationarity. By linearity and independence,

$$\mathbb{E}[y_t] \;=\; \sum_{j=0}^{\infty} \phi^j\, \mathbb{E}[\varepsilon_{t-j}] \;=\; 0,$$

$$\mathrm{Var}(y_t) \;=\; \sum_{j=0}^{\infty} \phi^{2j}\; \mathrm{Var}(\varepsilon_{t-j}) \;=\; \sigma^2 \sum_{j=0}^{\infty} \phi^{2j} \;=\; \frac{\sigma^2}{1 - \phi^2}.$$

Therefore the stationary marginal law is

$$y_t \;\sim\; \mathcal{N}\Big(\,0\,,\;\tfrac{\sigma^2}{1-\phi^2}\,\Big).$$

Invariance (consistency with the kernel). Let $v := \sigma^2/(1-\phi^2)$. Assume $y_{t-1} \sim \mathcal{N}(0, v)$ and use the transition $y_t = \phi y_{t-1} + \varepsilon_t$ with $\varepsilon_t \perp y_{t-1}$:

$$\begin{aligned}
\mathbb{E}[y_t] \;&=\; \phi\,\mathbb{E}[y_{t-1}] \;+\; \mathbb{E}[\varepsilon_t] \;=\; 0,\\
\mathrm{Var}(y_t) \;&=\; \phi^2\;\mathrm{Var}(y_{t-1}) \;+\; \mathrm{Var}(\varepsilon_t) \;=\; \phi^2 v \;+\; \sigma^2 \;=\; v.
\end{aligned}$$

Thus the Gaussian $\mathcal{N}(0, v)$ is invariant for the kernel $p(y_t \mid y_{t-1})$. Uniqueness follows from contractivity when $|\phi| < 1$.

(Useful corollary: autocovariance/ACF). For $h \geq 0$,

$$\gamma(h) \;=\; \mathrm{Cov}(y_t, y_{t-h}) \;=\; \phi^{\,h}\,\frac{\sigma^2}{1-\phi^2}, \qquad \rho(h) \;=\; \phi^{\,h}.$$

(a) Transition density: $p(y_t \mid y_{t-1}) = \dfrac{1}{\sqrt{2\pi\sigma^2}}\,\exp\Big\{-\tfrac{1}{2\sigma^2}\,(y_t - \phi y_{t-1})^2\Big\}$,

$\Rightarrow$ AR(1) is time–homogeneous Markov of order 1.

(b) Stationary law (if $|\phi| < 1$): $y_t \sim \mathcal{N}\Big(0,\;\tfrac{\sigma^2}{1-\phi^2}\Big)$, $\quad \gamma(h) = \phi^{\,h}\tfrac{\sigma^2}{1-\phi^2}$, $\quad \rho(h) = \phi^{\,h}$.

3. Nonlinear and Multi-Step Autoregressive Models

3.1 Nonlinear Autoregressive (NAR) Model

We model, for $t > p$,

$$p_\theta(x_t \mid x_{<t}) \;=\; \mathcal{N}\Big(f_\theta(x_{t-1}, \ldots, x_{t-p})\;,\;\sigma^2\Big).$$

The conditional (innovations) likelihood of a sequence $x_{1:T}$ given the first p values is

$$L(\theta \mid x_{1:T}) \;=\; \prod_{t=p+1}^{T} \frac{1}{\sqrt{2\pi\sigma^2}}\,\exp\left\{-\frac{1}{2\sigma^2}\,\Big(x_t - f_\theta(x_{t-1:t-p})\Big)^2\right\}.$$

Negative log-likelihood (NLL):

$$\begin{aligned}
\mathrm{NLL}(\theta) \;&=\; -\log L(\theta \mid x_{1:T})\\
&=\; \frac{T-p}{2}\,\log(2\pi\sigma^2) \;+\; \frac{1}{2\sigma^2}\sum_{t=p+1}^{T}\Big(x_t - f_\theta(x_{t-1:t-p})\Big)^2.
\end{aligned}$$

Gradients via BPTT (how to compute them). Let $e_t := x_t - f_\theta(\cdot)$. Then

$$\nabla_\theta \mathrm{NLL}(\theta) \;=\; -\frac{1}{\sigma^2}\sum_{t=p+1}^{T} e_t\,\nabla_\theta f_\theta(x_{t-1:t-p}).$$

Because f_θ is a neural net, $\nabla_\theta f_\theta$ is obtained by standard backprop *through* the network at each time step, and the whole sequence is handled with backpropagation-through-time (BPTT). If you unroll the computational graph over $t = p+1, \dots, T$, autodiff takes care of the chain rule.

Gradient flow in AR. Autoregression couples time steps only through inputs $x_{t-1:t-p}$, so gradients flow through at most p lags per step. Long horizons T or large p can still cause vanishing/exploding effects since BPTT multiplies Jacobians across the unrolled depth; regularization, clipping, and careful architecture (next subsection) help.

$$\mathrm{NLL}(\theta) = \frac{T-p}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_{t=p+1}^{T}\Big(x_t - f_\theta(x_{t-1:t-p})\Big)^2,$$

$$\nabla_\theta \mathrm{NLL}(\theta) = -\frac{1}{\sigma^2}\sum_{t=p+1}^{T}\Big(x_t - f_\theta(x_{t-1:t-p})\Big)\,\nabla_\theta f_\theta(x_{t-1:t-p}).$$

3.2 Architecture for Large vs. Small Context

If the model needs a *very long* memory ($p \gg 1$), we should pick architectures that can look far back without paying a quadratic cost and without killing gradients. If the context is *short*, we can keep it simple and fast—no need to bring a Transformer to a linear-regression fight.

Large p (very long context). Use receptive-field or memory-efficient designs:

(i) Dilated causal CNNs (e.g., WaveNet-style) for $O(T)$ compute,

(ii) State-space or linear RNNs (S4, DSS, Mamba) for long-range modeling,

(iii) Efficient causal Transformers (chunked, linear, or sparse attention),

(iv) Add residuals, normalization, clipping for stable gradients.

Small p (short context). Prefer compact learners:

(i) Small MLP on the lag vector $[x_{t-1}, \dots, x_{t-p}]$,

(ii) Shallow causal CNN with small kernels,

(iii) Lightweight GRU/LSTM if slight temporal state helps.

Large p : dilated causal CNNs, state-space/linear RNNs, efficient causal Transformers.
Small p : MLP on lags, shallow causal CNN, small GRU/LSTM.

3.3 Multi-Step Prediction

To forecast k steps ahead, we roll the model forward, feeding back predictions. This is where tiny one-step errors can snowball, because every new input may include yesterday's mistake. We can either harden the model to this exposure or make training mimic it.

Let $\widehat{x}_{t+1}$ be the one-step forecast using the latest p *available* values (true or predicted). Define the lag vector

$$\mathbf{z}_t := [\,x_t,\, x_{t-1},\, \dots,\, x_{t-p+1}\,]^\top.$$

Then

$$
\begin{aligned}
\widehat{x}_{t+1} &= f_\theta(\mathbf{z}_t),\\
\widehat{x}_{t+2} &= f_\theta([\,\widehat{x}_{t+1}, x_t, \ldots, x_{t-p+2}\,]),\\
&\vdots\\
\widehat{x}_{t+k} &= f_\theta([\,\widehat{x}_{t+k-1}, \widehat{x}_{t+k-2}, \ldots, \widehat{x}_{t+k-p}\,]).
\end{aligned}
$$

Why errors accumulate. If f_θ is L-Lipschitz in its inputs, a first-order bound gives

$$|\widehat{x}_{t+h} - x_{t+h}| \;\; L\,|\widehat{x}_{t+h-1} - x_{t+h-1}| \;+\; \text{model bias},$$

so the error can grow like $O(L^h)$. Linearization yields Jacobian products across steps, which is the same vanishing/exploding story as in deep nets.

Mitigations (training / evaluation).

(i) Scheduled sampling / professor forcing (expose model to its own outputs),

(ii) Multi-step losses $\displaystyle\sum_{h=1}^{k} \|x_{t+h} - \widehat{x}_{t+h}\|^2$,

(iii) Noise injection / data augmentation on inputs,

(iv) Closed-loop validation to match test-time rollouts.

$\widehat{x}_{t+h} = f_\theta([\widehat{x}_{t+h-1}, \ldots, \widehat{x}_{t+h-p}]), \quad h = 1, \ldots, k,$
Error growth (Lipschitz L): $|\widehat{x}_{t+h} - x_{t+h}| L^h\, |\widehat{x}_t - x_t| + \text{bias}$,
Mitigate via scheduled sampling, multi-step loss, noise, and closed-loop eval.

3.4 Conditional Nonlinear AR Modeling

Now the model gets an extra sequence $c_{1:T}$ (text, control, motion, etc.). The mean of the Gaussian at time t is allowed to depend on both the recent p values of x and the conditioning context up to t.

Model:

$$p_\theta(x_t \mid x_{<t}, c_{1:t}) \;=\; \mathcal{N}\Big(f_\theta(x_{t-1:t-p}, c_{1:t})\;,\; \sigma^2\Big).$$

Conditional likelihood of $x_{1:T}$ given $c_{1:T}$ (conditioning on initial p of x):

$$L(\theta \mid x_{1:T}, c_{1:T}) \;=\; \prod_{t=p+1}^{T} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{1}{2\sigma^2}\Big(x_t - f_\theta(x_{t-1:t-p}, c_{1:t})\Big)^2 \right\}.$$

Conditional log-likelihood and the training objective:

$$\log L(\theta \mid x, c) \;=\; -\frac{T-p}{2}\log(2\pi\sigma^2) \;-\; \frac{1}{2\sigma^2}\sum_{t=p+1}^{T}\Big(x_t - f_\theta(x_{t-1:t-p}, c_{1:t})\Big)^2,$$

$$\mathrm{NLL}_{\mathrm{cond}}(\theta) \;=\; -\log L(\theta \mid x, c) \;=\; \frac{T-p}{2}\log(2\pi\sigma^2) \;+\; \frac{1}{2\sigma^2}\sum_{t=p+1}^{T}\Big(x_t - f_\theta(\cdot)\Big)^2.$$

Use cases and differences. This lets you do conditional generation (e.g., speech from text, motion from control). Compared to the unconditional AR, the factorization is the same but every conditional mean may now attend to c, changing both capacity and the dependence structure (you can hold c fixed and resample x).

Final (3.4): Conditional objective.

$$p_\theta(x_{1:T} \mid c_{1:T}) = \prod_{t=p+1}^{T} \mathcal{N}\Big(x_t\,;\, f_\theta(x_{t-1:t-p}, c_{1:t}),\, \sigma^2\Big),$$

$$\mathrm{NLL}_{\mathrm{cond}}(\theta) = \frac{T-p}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_{t=p+1}^{T}\Big(x_t - f_\theta(x_{t-1:t-p}, c_{1:t})\Big)^2.$$

3.5 KL Divergence between Two Conditional NAR Models

What to keep in mind: We compare two conditional AR models with the *same variance* σ^2: one uses mean f_θ, the other g_ϕ. Because both factorize autoregressively, the sequence-level KL neatly sums the per-time KLs of the Gaussians. The expectation is under the first model's rollout, which we approximate by Monte Carlo.

Let

$$p_\theta(x_t \mid x_{<t}, c_{1:t}) = \mathcal{N}\Big(\mu_t^\theta,\ \sigma^2\Big), \quad \mu_t^\theta := f_\theta(x_{t-1:t-p}, c_{1:t}),$$
$$q_\phi(x_t \mid x_{<t}, c_{1:t}) = \mathcal{N}\Big(\mu_t^\phi,\ \sigma^2\Big), \quad \mu_t^\phi := g_\phi(x_{t-1:t-p}, c_{1:t}).$$

AR factorization (conditioned on initial p of x):

$$p_\theta(x_{1:T} \mid c) = \prod_{t=p+1}^{T} p_\theta(x_t \mid x_{<t}, c), \qquad q_\phi(x_{1:T} \mid c) = \prod_{t=p+1}^{T} q_\phi(x_t \mid x_{<t}, c).$$

Sequence-level KL:

$$\mathrm{KL}\big(p_\theta(\cdot \mid c) \,\|\, q_\phi(\cdot \mid c)\big) = \mathbb{E}_{p_\theta(x_{1:T}|c)}\left[\sum_{t=p+1}^{T} \log\frac{p_\theta(x_t \mid x_{<t}, c)}{q_\phi(x_t \mid x_{<t}, c)}\right].$$

For univariate Gaussians with the same variance σ^2,

$$\mathrm{KL}\big(\mathcal{N}(\mu_1, \sigma^2) \,\|\, \mathcal{N}(\mu_2, \sigma^2)\big) = \frac{1}{2\sigma^2}(\mu_1 - \mu_2)^2.$$

Therefore,

$$\mathrm{KL}\big(p_\theta(\cdot \mid c) \,\|\, q_\phi(\cdot \mid c)\big) = \frac{1}{2\sigma^2}\,\mathbb{E}_{p_\theta(x_{1:T}|c)}\left[\sum_{t=p+1}^{T}\Big(\mu_t^\theta - \mu_t^\phi\Big)^2\right].$$

Computation in practice (high dimension). The expectation over full rollouts of p_θ is intractable in closed form; sample S trajectories $x_{1:T}^{(s)} \sim p_\theta(\cdot \mid c)$ by ancestral sampling and approximate

$$\widehat{\mathrm{KL}} \ \approx\ \frac{1}{2\sigma^2 S}\sum_{s=1}^{S}\sum_{t=p+1}^{T}\Big(\mu_t^\theta(x_{<t}^{(s)}, c) - \mu_t^\phi(x_{<t}^{(s)}, c)\Big)^2.$$

If rollouts are unstable, use teacher-forced paths from data or a mixture (importance weighting) as a lower-variance surrogate.

$$\mathrm{KL}\big(p_\theta(\cdot \mid c) \,\|\, q_\phi(\cdot \mid c)\big) = \frac{1}{2\sigma^2}\, \mathbb{E}_{p_\theta(x_{1:T}|c)}\left[\sum_{t=p+1}^{T} \Big(f_\theta(x_{t-1:t-p}, c_{1:t}) - g_\phi(x_{t-1:t-p}, c_{1:t})\Big)^2\right],$$

$$\text{Monte Carlo:}\quad \widehat{\mathrm{KL}} = \frac{1}{2\sigma^2 S} \sum_{s=1}^{S} \sum_{t=p+1}^{T} \Big(\mu_t^\theta(x_{<t}^{(s)}, c) - \mu_t^\phi(x_{<t}^{(s)}, c)\Big)^2.$$

4. Real NADE Parameters

To state our proposition, we introduce a slightly different but equivalent notation for NADE parameters. The new mixture NADE model should be an extension of the original real-valued NADE. We can therefore reuse the same idea and introduce new parameters for the mixture components. In fact, each π_i^c, μ_i^c and σ_i^c corresponds to a weight vector and a bias.

Recalling the original modeling of NADE, the nonnegativity condition had to be satisfied for σ_i^c to be valid. This was enforced with the exponential function, which is always nonnegative; thus we can pass the parameter through the exponential to guarantee the constraint.

The same story exists for π_i^c except we now have two conditions that must be met:

- $\forall i, c:\ \pi_i^c \geq 0$
- $\forall i:\ \sum_{c=1}^{C} \pi_i^c = 1$

A standard function that satisfies both simultaneously is the *softmax*. We now formalize the parametrization of our mixture NADE model.

Solution.

$$\begin{aligned}
\mu_i^c &= (w_{i,c}^\mu)^\top h_i + b_{i,c}^\mu,\\
\sigma_i^c &= \exp\big((w_{i,c}^\sigma)^\top h_i + b_{i,c}^\sigma\big),\\
\pi_i^c &= \mathrm{Softmax}\big((v_{i,c}^\pi)^\top h_i + b_{i,c}^\pi\big) = \frac{\exp\big((v_{i,c}^\pi)^\top h_i + b_{i,c}^\pi\big)}{\sum_{k=1}^{C} \exp\big((v_{i,k}^\pi)^\top h_i + b_{i,k}^\pi\big)}.
\end{aligned}$$

It is time to count the number of parameters for the proposed model. An important consideration is whether we also count the parameters that define the hidden vector h_i. We first count *without* h_i's own parameters, i.e., for a single conditional $p(x_i \mid x_{<i})$.

Solution.

$$\begin{aligned}
&b_{i,c}^\mu \longrightarrow C, \qquad b_{i,c}^\sigma \longrightarrow C, \qquad b_{i,c}^\pi \longrightarrow C,\\
&v_{i,c}^\mu \longrightarrow dC, \qquad v_{i,c}^\sigma \longrightarrow dC, \qquad v_{i,c}^\pi \longrightarrow dC.
\end{aligned}$$

$$\begin{aligned}
\text{Number of parameters for } (i \neq 1) &= 3C(d+1),\\
\text{Number of parameters for } (i = 1) &= 3C.
\end{aligned}$$

Now include the parameters that define the hidden representation

$$h_i \;=\; \sigma\big(W_i x_{<i} \;+\; c_i\big), \qquad W_i \in \mathbb{R}^{d\times(i-1)}, \quad c_i \in \mathbb{R}^d.$$

The additional parameters are $(i-1)d$ entries for W_i and d entries for c_i, i.e., id in total (with the convention $i = 1 \Rightarrow 0$ for W_1).

Number of parameters for $(i \neq 1) \;=\; 3C(d+1) \;+\; id,$
Number of parameters for $(i = 1) \;=\; 3C \;+\; d.$

Summing over all $i = 1, \ldots, n$ we obtain the total parameter count for the full model.

Total number of parameters $\;=\; 3C(d+1)(n-1) \;+\; 3C \;+\; nd.$

Autoregressive Graph Neural Networks for Sequential Data

We have a multivariate time series with d variables. Each variable is treated as a node v in a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

At each time t and node v, the hidden state is updated by a GNN:

$$h_t^v = \mathrm{GNN}\big(h_{t-1}^v, \{h_{t-1}^u : u \in N(v)\}\big).$$

This just means: to update node v at time t, we look at its own hidden state at time $t-1$ and the hidden states of its neighbors $N(v)$ at time $t-1$ and run them through some shared GNN function.

We observe values x_t^v at each node and time.

2. Missing node values at test time.

At test time, some nodes might be missing (sigh, this always happens in real data).

For each node v and time t we have a binary mask $m_t^v \in \{0, 1\}$:

$$m_t^v = 1 \Rightarrow \text{node } v \text{ is observed at time } t, \qquad m_t^v = 0 \Rightarrow \text{node } v \text{ is missing at time } t.$$

We define the observed set and missing set as

$$\mathcal{O}_t = \{v : m_t^v = 1\}, \qquad \mathcal{M}_t = \{v : m_t^v = 0\}.$$

(a) Training objective with masking.

During training we only know the true value x_t^v for $v \in \mathcal{O}_t$. So our log-likelihood can only sum over those nodes.

Let p_θ be our model (the GNN) with parameters θ. We define the loss:

$$\mathcal{L}(\theta) = -\sum_{t=1}^{T} \sum_{v \in \mathcal{O}_t} \log p_\theta\big(x_t^v \,\big|\, \tilde{x}_{<t}, m_{<t}\big).$$

A few words what each symbol is:

- $p_\theta(x_t^v \mid \tilde{x}_{<t}, m_{<t})$ is the predicted probability (or density) of x_t^v, given the past inputs and masks.

- $\tilde{x}_{<t}$ are the input node features for times $1, \ldots, t-1$, where *missing* values have been filled with some default value (like zero, or maybe last observed value, or some learned embedding).

- $m_{<t}$ are the masks for all previous times, and we feed them into the GNN as extra features so the network knows "this value was missing, don't trust it too much".

So the training objective is just: *sum the log-probabilities over all the observed nodes and timesteps, and try to maximize that (or equivalently minimize $-\mathcal{L}$).*

To make the model more robust to missing stuff, we can also apply **random masking** during training: even when a node is actually observed, we sometimes pretend it is missing (drop it out) with some probability. Concretely, we can sample an extra mask $\hat{m}_t^v \in \{0, 1\}$ and use the combined mask

$$\bar{m}_t^v = m_t^v \wedge \hat{m}_t^v,$$

and only include nodes with $\bar{m}_t^v = 1$ in the loss. This forces the GNN to learn to rely on neighbors and time structure, not just always the same full set of inputs.

(b) Inferring missing node values at test time.

After the model is trained, at test time we wanna fill in the missing nodes x_t^v for $v \in \mathcal{M}_t$.

The GNN gives us a conditional distribution for each node at time t:

$$p_\theta\big(x_t^v \mid x_{<t}^v, x_{<t}^{N(v)}, m_{\le t}\big),$$

where $m_{\le t}$ are the masks up to time t. The dependence on hidden states is implicit here, but it's encoded by the GNN through the autoregressive updates.

The procedure is roughly:

1. Feed the sequence (with missing entries replaced by some placeholder) and the masks into the GNN to get hidden states h_t^v for all t, v.

2. For every missing node $v \in \mathcal{M}_t$, the model gives the conditional $p_\theta(x_t^v \mid \cdots)$ as above. We can then

   - either use the *mean* (or mode) of this distribution as a point estimate:

   $$\hat{x}_t^v = \mathbb{E}_{p_\theta}\big[x_t^v \mid x_{<t}^v, x_{<t}^{N(v)}, m_{\le t}\big],$$

   - or we can *sample* from it:

   $$\hat{x}_t^v \sim p_\theta\big(x_t^v \mid x_{<t}^v, x_{<t}^{N(v)}, m_{\le t}\big),$$

   if we want stochastic generation or multiple imputations.

So, in plain words: *we run the GNN forward using whatever we have, let it tell us the distribution of each missing node given its history and neighbors, and then we either take the average of that distribution or sample from it to "fill in" the blanks.*

This way the same conditional distributions we trained with are used both to model the data and to impute the missing values at test time.